6. Proxy Pattern.

Provide a surrogate or placeholder for another object to controll access to it.

In simple terms:-

- Proxy acts as a middle layer.
- Client interacts with proxy instead of the real object.
- Proxy decides how and when to access the real object.

* Why Proxy Exists

sometimes accessing an object is:-

- Expensive
- Sensitive
- Remote
- Slow.

Example

- Large image loading
- DB Connection
- Network service.
- Access - controlled service.

Proxy helps by:-

- Delaying creation
- Controlling access
- Adds extra logic.

Instead of Client → Real object.

we do Client → Proxy → Real Object.

The proxy:

- Implements the same interface
- Controls access of the real object.

Ex:- Security Guard.

You don't enter a restricted building directly.

you first intract with:

Security Guard → verify access → allow entry.

Security Guard = Proxy.

Code

1. Subject Interface.

```
interface Image{
    void display();
}
```

2. Real object

```
class RealImage implements Image{
  private String filename;
   RealImage (String Filename){
     this.filename= Filename;
     loadFromDisk();
   }
  Private void loadFromDisk(){
    s.o.pln ("Loading "+ filename);
  }
  Public void display(){
   s.o.pln (" Displaying"+ Filename);
  }
}
```

3. Proxy class

```
class ImageProxy implements Image{
  Private RealImage realImage;
  private String Filename;

  ImageProxy (String filename){
     this.filename= Filename;
  }
  public void display(){
   if (realImage ==null){
       realImage= new RealImage (Filename);
   }
   realImage.display();
}}
```

4. Client Code.

```
Image image = new ImageProxy ("photo.png");
image.display();
image.display();
```

- lazy loading
- Controlled access
- performance optimization.

Types of Proxy

1. Virtual Proxy — used for lazy loading.
   Eg. • loading large Images.
   • Lazy initilization.

2. Protection proxy — controlled access permissions.
   • user roles.
   • security checks.

3. Remote proxy. — represents an obj in another system.
   • RPC.
   • REST clients.

4. Caching Proxy. — shows results to improve performanie.
   • API response caching.